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LEARNER WORKBOOK

DEMONSTRATE AN UNDERSTANDING OF DIFFERENT COMPUTER NETWORK ARCHITECTURES AND STANDARDS

UNIT STANDARD 114074

NQF LEVEL: 5

CREDITS: 5

NOTIONAL HOURS: 50

LEARNER INFORMATION	
Name	
Surname	
ID Number	
Contact	

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OVERVIEW

About the Learner Workbook	This Learner Exercise Workbook has been designed and developed to evaluate learners' level of understanding It forms part of a series of Learner Workbooks that have been developed for **NATIONAL CERTIFICATE: INFORMATION TECHNOLOGY: SYSTEMS SUPPORT:SAQA 48573 -LEVEL 5- 147 CREDITS**		
Purpose	The purpose of this Learner Exercise Workbook is to evaluate learners understanding on the specific outcomes and/or assessment criteria of the following SAQA Registered Unit Standards:		
US No	**US Title**	**Level**	**Credits**
	Saqa 114074 **DEMONSTRATE AN UNDERSTANDING OF DIFFERENT COMPUTER NETWORK ARCHITECTURES AND STANDARDS**		
Context	This assessment represents the Formative Assessment component of Function in a business environment,for the should be completed in the classroom/training room.		
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.		
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the workbook to each learner; and • Interview the learner on similar questions, should he/she not be able to write.		

Instructions to Learners	Learners will be required to: • Complete the workbook as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame of…. hours.
Total Mark	This formative assessment carries a total mark of _ **points**. In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.
Equipment	Learners are required to have the following equipment in order to complete this workbook: • Pen and Pencil; • Ruler; and • Exam Pad – for additional paper.

GENERAL INFORMATION

LEARNER DETAILS		
Learner Full Names		
Learner ID No.:		
Organisation:		
Unit/Dept:		
Contact Details:	**Telephone /Cell Numbers:**	**Email Address:**
WORKSHOP DETAILS		
Workshop Venue:		
Facilitator Name:		
Date Started:		
Date Completed:		

ASSESSMENT PREPARATION CHECKLIST

DESCRIPTION	YES	NO	COMMENTS/CONTINGENCY
This assessment is a formative assessment and it is based on the outlined unit standard/s for module.			
Your assessment evidence for module needs to be submitted on..... (day) of.....(month).....(year) at the following address/place.....			
You will be assessed based on the outlined Unit Standards. The assessment activities are linked to specific outcomes/assessment criteria of the outlined Unit Standards.			
To determine your competence level, the following are the methods to be used for this assessment: 1.			

2.			
To be declared competent on module (formative assessment), you should have obtained at least 80% of the total mark of this assessment.			
You will be provided with detailed feedback on your performance of this assessment as follows: 1. Written Feedback 2. Verbal Feedback			
Should you be declared "not yet competent" on this assessment, you will be entitled for re-assessment opportunity/ies.			
You will be required to re-submit evidence (only for areas) you were declared not yet competent. A date for re-submission will be agreed with the assessor.			
You will be entitled to lodge an appeal should you not be satisfied with the assessment decision of your assessment.			
You will be required to provide the assessor feedback on assessment procedure – this is to assist in improving the assessment practices.			
Your results of assessment and portfolio of evidence information will not be provided to any person without your written consent.			

Learner's Declaration			
I.....herewith declare that I am ready for the assessment, that we have reviewed the assessment preparation and plan, I understand the assessment process and I am happy that the assessment will be conducted in a fair manner.			
Learner Signature:	**Date:**	**Facilitator Signature:**	**Date:**

Demonstrate an understanding of different computer network architectures and standards

Learning Unit 1

UNIT STANDARD NUMBER	:	114074
LEVEL ON THE NQF	:	5
CREDITS	:	5
FIELD	:	Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:	• This unit standard is intended : • To provide fundamental knowledge of the areas covered • For those entering the workplace in the area of Data Communications & Networking. • As additional knowledge for those wanting to understand the areas covered • People credited with this unit standard are able to: • Describe network types and standards • Explain local and wide area network architectures • Explain wide area network (WAN) architectures • The performance of all elements is to a standard that allows for further learning in this area.
LEARNING ASSUMED TO BE IN PLACE:	
The credit value of this unit is based on a person having prior knowledge and skills to: Demonstrate an understanding of fundamental mathematics (at least NQF level 3). Demonstrate PC competency skills (End-User Computing unit Standards, at least up to NQF level 3.). Describe data communications.	

SPECIFIC OUTCOME 1.

Describe computer network types and standards.

Learning Outcomes

- 1. The description distinguishes types of networks.
- 2. The description compares network topologies.
- 3. The description outlines and distinguishes features of node addressing methods.
- 4. The description outlines the standards for industry network architectures.

Activity	Questions Description	Mark
1	• **Describe computer network types and standards**	10

SPECIFIC OUTCOME 2.

The explanation outlines channel utilisation techniques and their features.

Learning Outcomes

- The explanation outlines channel utilisation techniques and their features.
- The explanation compares the principles of access protocols.
- The explanation identifies the purpose of LAN components.
- The explanation illustrates network configurations and installation issues.

Activity	Questions Description	Mark
2	**There exist two LAN transmission options, Baseband and Broadband. Distinguish between the two**	10

SPECIFIC OUTCOME 3.

Explain LAN architectures.

Learning Outcomes

- The explanation identifies switching techniques.
- The explanation compares switching techniques.
- The explanation outlines WAN management issues.

Activity	Questions Description	Mark
3	**Explain LAN architectures.**	**10**

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				
7				

_____ _____

Learner Signature **Date**

_____ _____

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met** ☐	**Requirements not met** ☐
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER	
I, …….declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.	
Learner Date	**Facilitator Date**

DECLARATION BY THE ASSESSOR	
I ……..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.	
Assessor Date	**Moderator Date**